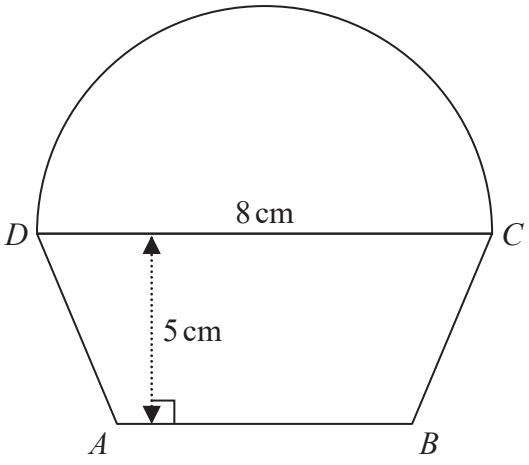
24 The diagram shows a shape made from a trapezium $ABCD$ and a semicircle with diameter DC.



$DC = 8$ cm
The shape has area 64 cm^2
The height of the trapezium is 5 cm.

Work out the length of AB.
Give your answer correct to 1 decimal place.

Semicircle: $r = 4$, area $= \frac{1}{2}\pi \times 4^2 = 8\pi$

1 mark
Find semicircle area.

Trapezium area $= 64 - 8\pi = 38.8672...$

1 mark
Subtract from total area.

$A = \frac{1}{2}(a + b)h$

$38.8672... = \frac{1}{2}(AB + 8) \times 5$

1 mark
Use trapezium area formula.

$AB + 8 = 2(64 - 8\pi) \div 5 = 15.5469...$

$AB = 15.5469... - 8 = 7.5469...$

1 mark
Subtract 8.

$AB = 7.5$ cm (1 decimal place)

1 mark
Round at the end.

..... cm

(Total for Question 24 is 5 marks)